

Transparent Assignment Design Worksheet

PURPOSE

Skills: What skills will participants practice when doing the activity (both disciplinary skills and transferable skills)? **Disciplinary skills:** Application of OR framework
Critically evaluating cases from multiple perspectives
Analyzing relevant trade-offs from multiple perspectives
Reflective writing
Transferable skills: Critical thinking
Make sequential decisions based on facts and arguments
Recognizing and weighing consequences of decisions
Communication abilities: oral and written
Applying learned concepts to practical situations
Knowledge: What knowledge will participants gain?
Empathize with or understand different stakeholders (perspectives) of the same problem
Confidence in decision-making
Understanding of professional responsibility (in decision-making)
Long term relevance: What connections are there between these skills/knowledge and the students' own lives?
Their career will be to find appropriate solutions to real-world problems, most commonly than not, involving different stakeholders.
The coursework in IE is mainly problem-solving and modeling, both focus of this activity.
Now turn these ideas into a student-facing purpose statement.
The purpose of this assignment is to translate a practical problem in public programs into a mathematical modeling structure. You will need to justify their choices regarding their modeling decisions regarding the stakeholder and problem.
This will help you later in the career when you must propose effective solutions to different problems.
As a result of completing this assignment, you will:
Increase your familiarity with OR tools and framework
Be able to evaluate trade-offs of different modeling decisions
Evaluate how the decision might change depending on the stakeholder

TASK

What – Very simply, what will participants do?- Students will discuss among their peers (in groups of min. 3 students) the assumptions of the case given, and define what the possible decisions are, and who are the decision-makers and stakeholders that are not decision-makers.

How – What procedure or sequence will they use to go about the task?

Read material

Discussion with peers

List of assumptions and list of decisions to be made

Decision regarding what assumptions and decisions

- Report and presentation that demonstrate they understood the:

- problem

- different perspectives

- decision-makers

- perspectives chosen

- perspectives not considered

- assumptions considered

- Students will need to justify their choices regarding their modeling decisions, explaining why they opted for their specific approach and why they did not choose at least two others

- Justification of each choice based on stakeholders and assumptions made.

- Explain why you opted for a specific approach and why you did not choose at least another alternative

Now turn these ideas into a student-facing task description, with step-by-step instructions.

Read the case study material. Participate in discussion forum. (Week 2)

In groups of 3 or 4, discuss assumptions of the case given, and define what the possible decisions are, and who are the decision-makers and stakeholders that are not decision-makers. (Week 3)

Create *personas* for each of the stakeholders (at least 3), and create a dialogue between at least 2 opposing perspectives, while they are defending what they need and/or what they can do about it. (Week 3)

Part of the discussion will be included in the introduction of the report, and presentation as a sketch.

Write an introduction to the final report for instructor feedback. (Week 4)

Prepare a presentation that explain Problem, Stakeholders, Decision-Makers. The presentation will be subject to

Peer-feedback at the discussion forum. (Week 4)

Justify your choices regarding every decision made.

Analyze trade-offs of decisions

Use feedback to improve decision and report/presentation (Week 5)

Report on the assumptions chosen and how they affect which stakeholder (at least 2 opposing stakeholders) (Week 5)

9. Report and presentation need to clearly demonstrate (Week 5):

- what is the problem you are trying to solve

- what are the different stakeholders and what they must gain or lose

- what is the possible decision-makers and which one you chose to model and WHY

- what perspectives have you chosen to include in the model and WHY

- what are the perspectives you have not considered in the model and WHY

- The final model assumptions considered

CRITERIA for SUCCESS

If you do not already have a checklist of expectations or a rubric for this activity, or rubric in advance so students can self-evaluate. Share this with participants before the activity.

Criteria: What specific aspects of this work distinguish insufficient work from work that is satisfactory, or even excellent?

Personas: Excellent - create

Stakeholders' perspective: Excellent- identify most of the stakeholders and appropriately assign their demands and capabilities.

Decision-Makers: Identify all possible decision-makers and select appropriate decision-makers(s) for the problem and resources available.

Assumptions: Identify the main assumptions for the case and make appropriate selection based on stakeholders and decision-makers decisions.

Weighting: What features are non-negotiable and which are "nice to have"?

Must have:

At least 2 opposing stakeholder's perspective considered with justification of why or not chosen to be included in the final decision.

Discussion with relevant needs and capabilities among stakeholders' *personas*.

Nice to have:

Relatable or realistic personas discussion

Now turn this into a student-facing rubric or checklist.

Excellent — Satisfactory — Needs Improvement

Use of OR Framework — Includes all steps and steps are accurate (15) — Includes most of the steps and steps are not followed perfectly (10) — Does not follow steps or steps are followed incorrectly (5)

Stakeholders' perspective — Identify all main stakeholders for the case and create relatable personas. Appropriately assign their demands and capabilities. (30) — Identify all main stakeholders for the case but do not create relatable personas. Do not assign the demands and capabilities correctly to some stakeholders. (20) — Do not identify all main stakeholders and do not create personas. Do not assign the demands and capabilities correctly to most of the stakeholders (7)

Decision-Makers — Identify all possible decision-makers and appropriately justify which one(s) will be considered for the problem and what resources are available. (25) — Identify most of the possible decision-makers and select appropriate but incomplete decision-makers(s) for the problem and resources available. (18) — Poorly identify the possible decision-makers and poorly select appropriate decision-makers(s) for the problem and resources available. (5)

Assumptions — Identify the main assumptions for the case and make appropriate selection based on the stakeholders and decision-makers' decisions. (30) — Identify most of the main assumptions for the case and do not make justification for the selection of some based on the stakeholders and decision-makers' decisions. (20) — Inappropriate choice of assumptions based on the stakeholders and decision-makers' decisions. Some main assumptions were not considered. (7)

MODELS of SUCCESS

Examples: Where might you find a range of examples of excellent work to show students what success looks like? Examples of case studies that are not being used in the class, and how they were analyzed (report and presentation format). There will be a blank checklist so they can analyze it for themselves and a filled one with annotations.

Applying the criteria: How might it be possible for participants to apply your checklist or rubric to an imperfect example of work before they are asked to produce their own?

An imperfect example will be showed and they will be provided with a blank checklist and a discussion post so they can share their "grading" and their opinion about the problem.